PRACTICAL SWINE ANESTHESIA AND CASTRATION TECHNIQUES

Yu Pin Huan

OBJECTIVE

- Practical approach of pig anesthesia
  - For lab environment
  - For farm management
  - For pet pig
- Pigs are not as tough as you think

PROBLEMS, PECULIARITIES AND PRACTICALITIES

- Body weight
  - Variety in size
  - Inexperience
  - Accurate measurement
- Handling
  - Nothing to hold on to
  - Strong
  - Reluctant to be handled
  - Pig board
  - Snare
  - Squeal





PROBLEMS, PECULIARITIES AND PRACTICALITIES

- Drug administration
  - 4 cm subcutaneous fat
    - Underestimate body weight
    - Injection into fat tissue
      - Poor blood supply
  - 18 gauge, 1 ·5 inch (3 ·75 cm) needle
  - Neck muscle





PROBLEMS, PECULIARITIES AND PRACTICALITIES

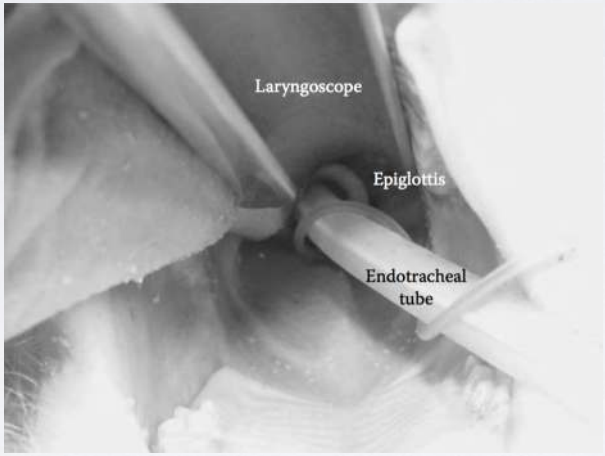
- IV port
  - Not very accessible
  - Auricular vein

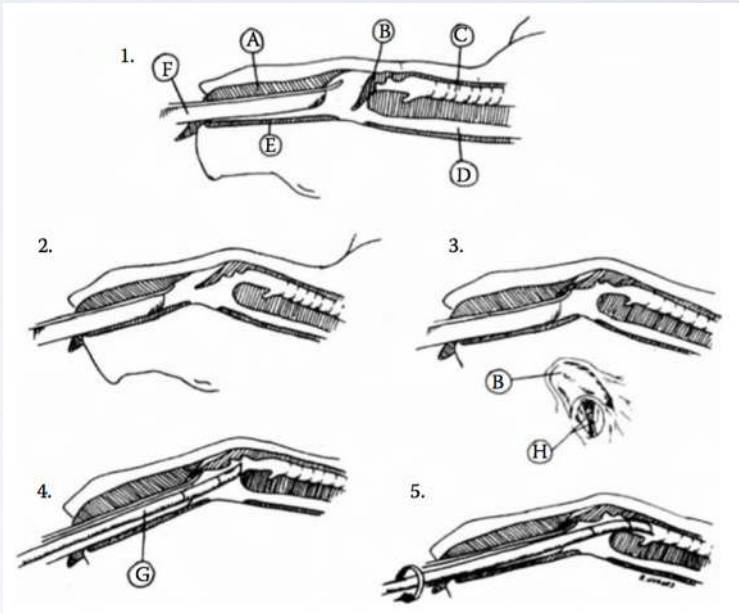




PROBLEMS, PECULIARITIES AND PRACTICALITIES

- Trachea intubation
  - Narrow glottis
  - Long soft plate
  - Fat cheek
  - Narrow jaw angle





  - Dorsal recumbency usually allows the excessive pharyngeal tissue to 'fall away' from the glottis, thus providing a good airway

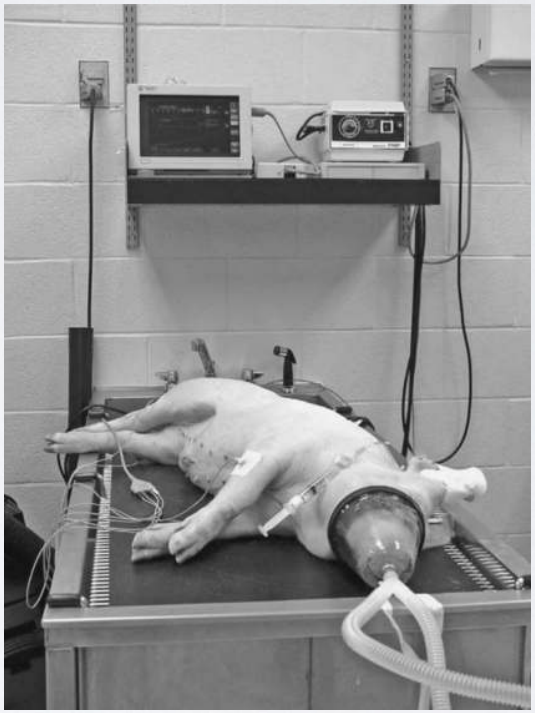
PROBLEMS, PECULIARITIES AND PRACTICALITIES

- Thermoregulation
  - Hypothermia
    - Lots of subcutaneous fat
    - Little hair
    - Four limbs
    - Young patient
    - >36 ◦ c
- Fasting
  - Vomiting
  - At least 6 hours
  - Ideally 12 hours



Sedation and premedication

- Indication
  - Minor procedures
- Brachycephalic breeds
  - Respiratory obstruction



Sedation and premedication

- Phenothiazine
  - Acepromazine
    - Vasodilation
      - Risk of hypothermia
- Benzodiazepam
  - Use in combination with ketamine
- Alpha 2-agonist
  - Not as potent as in ruminant
  - Vomiting

SEDATIVE AGENTS COMMONLY USED IN THE PIG

Agent	*Dose rate*	*Route of administration*
Azaperone	1 to 8 mg/kg	Deep intramuscular
Azaperone and ketamine	1 mg/kg 5 mg/kg	Together deep intramuscular
Azaperone, ketamine ± butorphanol	1 to 2 mg/kg 5 mg/kg 0·1 to 0·2 mg/kg	All together deep intramuscular
Acepromazine	0·03 to 0·1 mg/kg	Deep intramuscular
Acepromazine and ketamine	0·03 to 0·05 mg/kg 5 mg/kg	Together deep intramuscular
Midazolam	0·1 to 0·5 mg/kg	Deep intramuscular or intravenous
Midazolam and ketamine	0·2 mg/kg 5 mg/kg	Together deep intramuscular
Diazepam	0·5 to 2 mg/kg 1 to 10 mg/kg	Intravenous Deep intramuscular
Diazepam and ketamine	2 mg/kg 5 to 10 mg/kg	Intravenous followed by intravenous to effect
Xylazine, ketamine ± butorphanol	1 to 3 mg/kg 5 mg/kg 0·1 to 0·2 mg/kg	Together deep intramuscular

Note that none of the drugs listed above are licensed for use in pigs. An agent can only be used if it appears in one of the annexes (I to IV) of the Council Regulation (EEC) 2377/90

Injectable anesthetics

- Ketamine
  - Analgesia
  - Minimal cardiovascular effect
  - Excessive salivation
  - Rough recovery

INJECTABLE ANAESTHETICS COMMONLY USED IN THE PIG

Agent	*Dose*	*Method of administration*	*Analgesic effect*	*Cardiovascular depression*	*Respiratory depression*
Ketamine	10 to 20 mg/kg	Intramuscular and intravenous	Good	Minimal	Transient on induction
Thiopental	5 to 15 mg/kg	Intravenous	Minimal	Yes	Yes
Pentobarbital	5 to 15 mg/kg	Intravenous	Minimal	Mild	Mild
Alfadolone/alfaxalone	2 to 6 mg/kg	Intravenous and intramuscular	Some	Mild	Mild
Propofol	6 to 8 mg/kg	Intravenous	No	Yes	Yes

Injectable anesthetics

- Alphaxalone
  - Minor procedure
    - 15 min anesthesia
  - Cardiovascular depression
  - Minimal respiratory effect

Injectable anesthetics

- Propofol
  - IV route
  - Respiratory depression
  - Intubation necessary

Inhalational anesthetics

- Halothane
- Isoflurane
- Genetic
  - Miniature breeds

Malignant hyperthermia – a rare complication

Clinical signs

- Muscle rigidity, often leading to toe separation
- Retraction of the eyeballs far into the sockets due to increased extraocular muscle tone
- Very hot, pink blotchy skin
- Very pink mucous membranes
- Tachypnoea
- Tachycardia
- Increased temperature
- If monitoring with capnography, a rapidly increasing end-tidal carbon dioxide level

Treatment

Treatment can be difficult once clinical signs are observed, but involves:

- Removing the volatile agent and changing the circuit so that no residues enter the patient
- Rapid ventilation
- Administering cool fluids intravenously
- Performing colon lavage (ie, hosepipe in rectum)
- Placing ice packs and alcohol on the skin
- Administering dantrolene at 2 to 10 mg/kg intravenously

Monitor

- Cardiac monitoring
  - Susceptible to arrhythmia
    - Alpha2-agnoists, halothane
    - Farm animal > miniature breeds
    - ECG
    - Blood gas
    - Pulse oximetry

TABLE 2.1
Cardiopulmonary Emergency Drugs

Agent	Dosage	Indication
Aminophylline	5.0 mg/kg i.v.	Produces bronchodilation
Amiodarone	10.0–12.0 mg/kg, followed by 0.5–3.5 mg $kg^{-1}h^{-1}$ i.v.	Antiarrhythmic
Atropine	0.05 mg/kg i.v.	Counteracts bradycardia, heart block
Bicarbonate Na	1.0 mEq/kg bolus, followed by 0.5–1.0 mEq $kg^{-1}h^{-1}$	Counteracts acidosis
Bretylium	3.0–5.0 mg/kg i.v.	Antiarrhythmic
Calcium chloride	5.0–7.0 mg/kg slow i.v. infusion	Increases contractility
Digoxin	0.01–0.04 mg/kg i.v.	Counteracts supraventricular arrhythmias; decreases conduction; increases contractility
Dopamine	2.0–20.0 µg $kg^{-1}min^{-1}$ i.v.	Counteracts hypotension, cardiogenic shock
Dobutamine	2.5–10.0 µg $kg^{-1}min^{-1}$ i.v.	Counteracts hypotension, cardiogenic shock
Epinephrine	0.5–2.0 ml of 1:10,000 solution i.v. or i.c. (30 µg/kg)	Counteracts asystole, decreased contractility
Isoproterenol	0.01 µg $kg^{-1}min^{-1}$ i.v.	Induces brochodilation; counteracts AV block sinus bradycardia
Lidocaine	2.0–4.0 mg/kg bolus followed by 50 µg $kg^{-1}min^{-1}$ i.v.	Antiarrhythmic, antiectopic
Nitroprusside Na	0.5–0.8 µg $kg^{-1}min^{-1}$ i.v.	Reduces hypertension
Neosynepherine	0.5–1.0 mg/kg i.v.	Increases blood pressure by vasoconstriction
Propanolol	0.04–0.06 mg/kg i.v.	Counteracts tachycardia

Monitor

- Anesthetic depth
  - Heart rate, blood pressure
  - Muscle reflex
    - Jaw tone most reliable
  - Ocular reflex

Monitor

- Ventilation rate
  - Sensitive to over ventilation
    - Rupture
    - Emphysema
    - Pneumothorax
    - Ventilatory pressure
      - 18–22 cm H_2O.
    - Tidal volume
      - 5–10 ml/kg
    - End tidal CO_2
      - 40–50 mmHg
    - 12-15 breath/minute



Monitor



- Postoperative
  - HR, Body temperature…
    - Every 15 minutes
    - Until recovery of righting reflex

Pain management

- Little attention
  - Young piglets
    - Minimal distress
  - Assessment of analgesic effect
    - Variable

NSAIDs LICENSED FOR USE IN PIGS

	Dose	*Route of administration*
Meloxicam	0·4 mg/kg	Intramuscular injection
Ketoprofen	3 mg/kg	Deep intramuscular injection

Anesthetic agents and complications in Vietnamese potbellied pigs: 27 cases (1999–2006)

Cynthia M. Trim, BVSc, DACVA, and Christina Braun, Dr Med Vet, DACVA

- Premed
  - Butorphanol 0.18 mg/kg
  - Atropine 0.025 mg/kg
  - Midazolam 0.2 mg/kg

  - Xylazine 0.6 mg/kg or medetomedine 0.01 mg/kg
  - Butorphanol
  - Atropine

  - Zoletil 3.7 mg/kg
  - Butorphanol
- Induction
  - Ketamin
  - Propofol

COMPLICATIONS

- Pigs
  - Hypovetilation 67%
  - Hypotension 64%
  - Hypothemia 48%
  - Bradycardia 28%
  - Prolong recovery 22%

J Am Anim Hosp Assoc. 1999 Jan-Feb;35(1):13-7.

Complications and mortality associated with anesthesia in dogs and cats.

Gaynor JS[1], Dunlop CI, Wagner AE, Wertz EM, Golden AE, Demme WC.

- Complication
  - Hypotension (7%) dogs and 58 (8.5%) cats
  - Cardiac dysrhythmias(2.5%) dogs and 12 (1.8%) cats
  - Hypercapnea occurred (1.3%) dogs and (less than 1%) cat
  - Hypoxemia occurred in 14 (0.5%) dogs.

- Hypoventilation, hypotension, hypothermia were not associate with injectable anesthetic agent

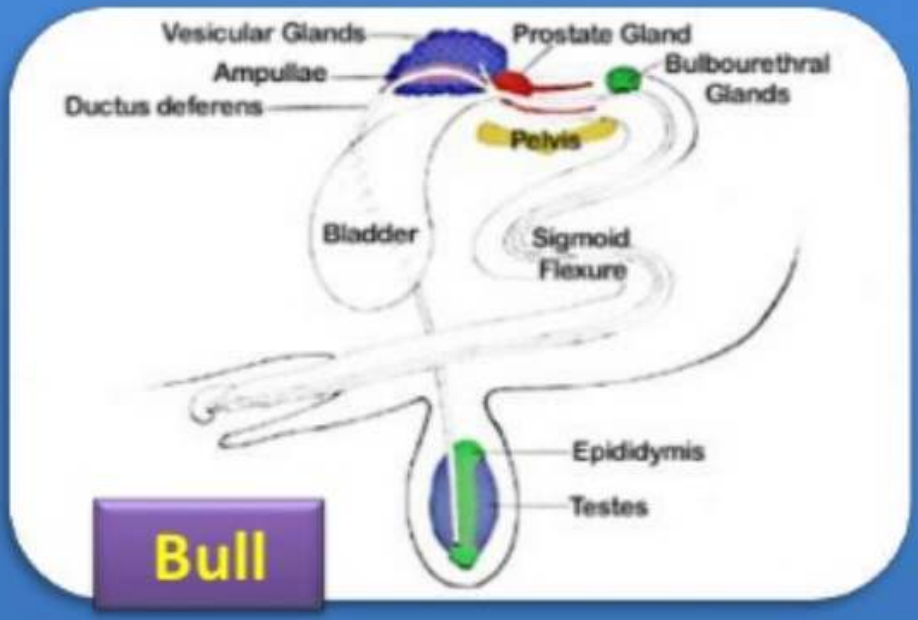
COMPLICATION

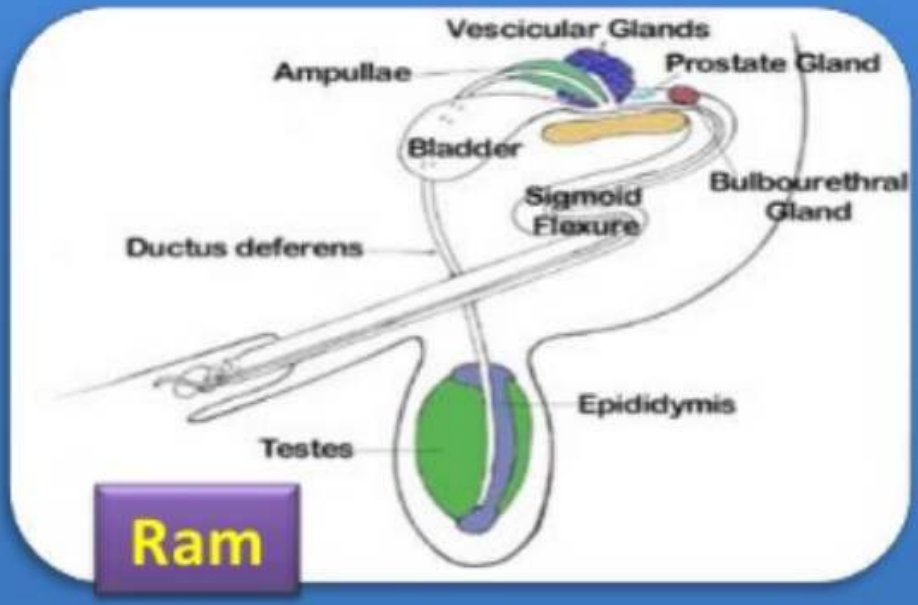
- Hypoventilation
  - Compression of muscle and fat
  - Absent of laryngeal adduction (potbellied pigs)
- Apnea after intubation
  - Effect of propofol (?)
  - Larnygospasm: easy induce in pigs
    - Apnea in respond to laryngeal stimulation had been reported in cats
- Heavy secretion
  - Easy in pigs
  - Atropine 0.02-0.04 mg/kg, glycopyrrolate 0.005-0.01 mg/kg

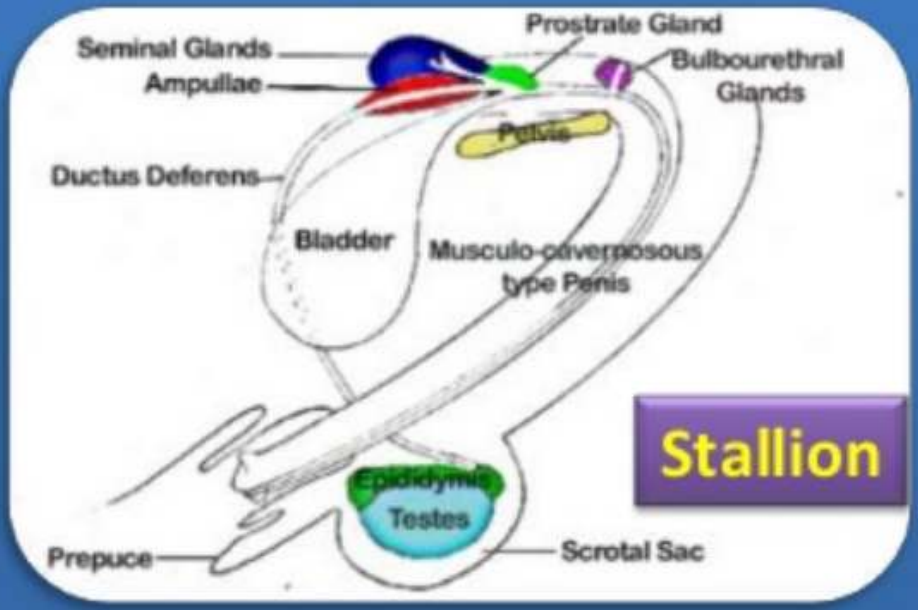
CONCLUSION

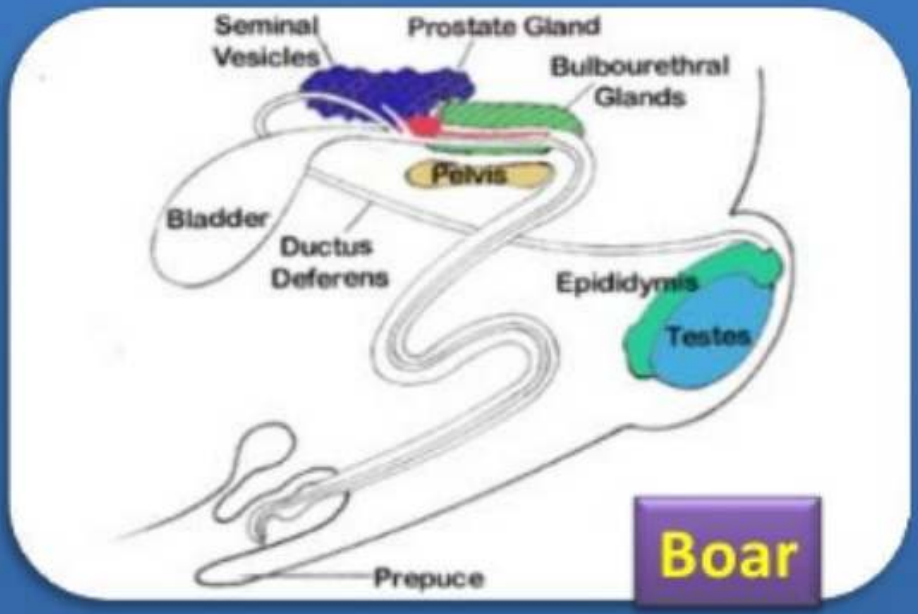
- Balanced anesthesia is required
- Pig is not as tough as you think
  - High complication rate
  - Most clinical environments are not ideal
- More information about analgesics and anesthetics is necessary

Male Reproductive System – Anatomy









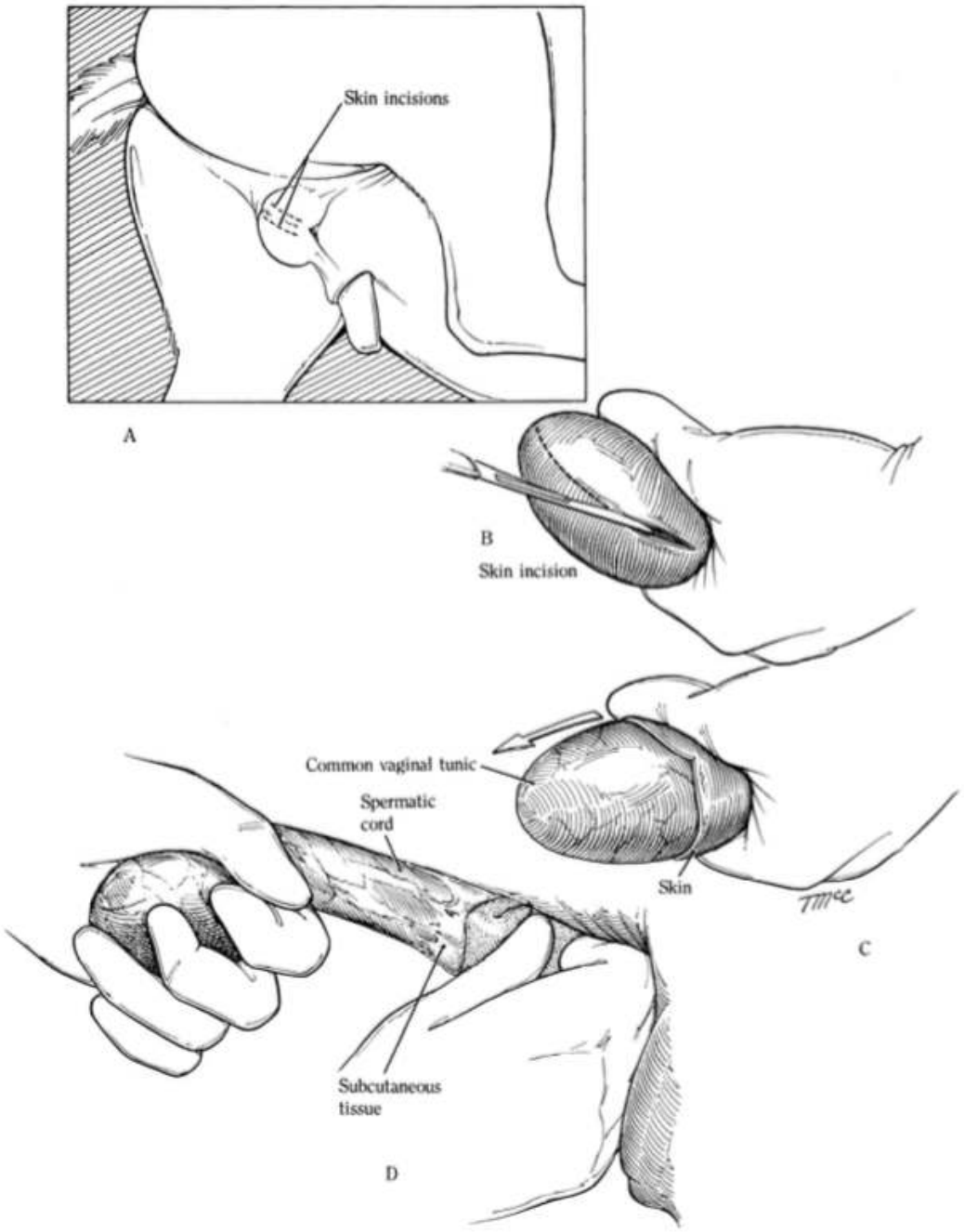


Fig. 10.1. **A–L.** Castration.

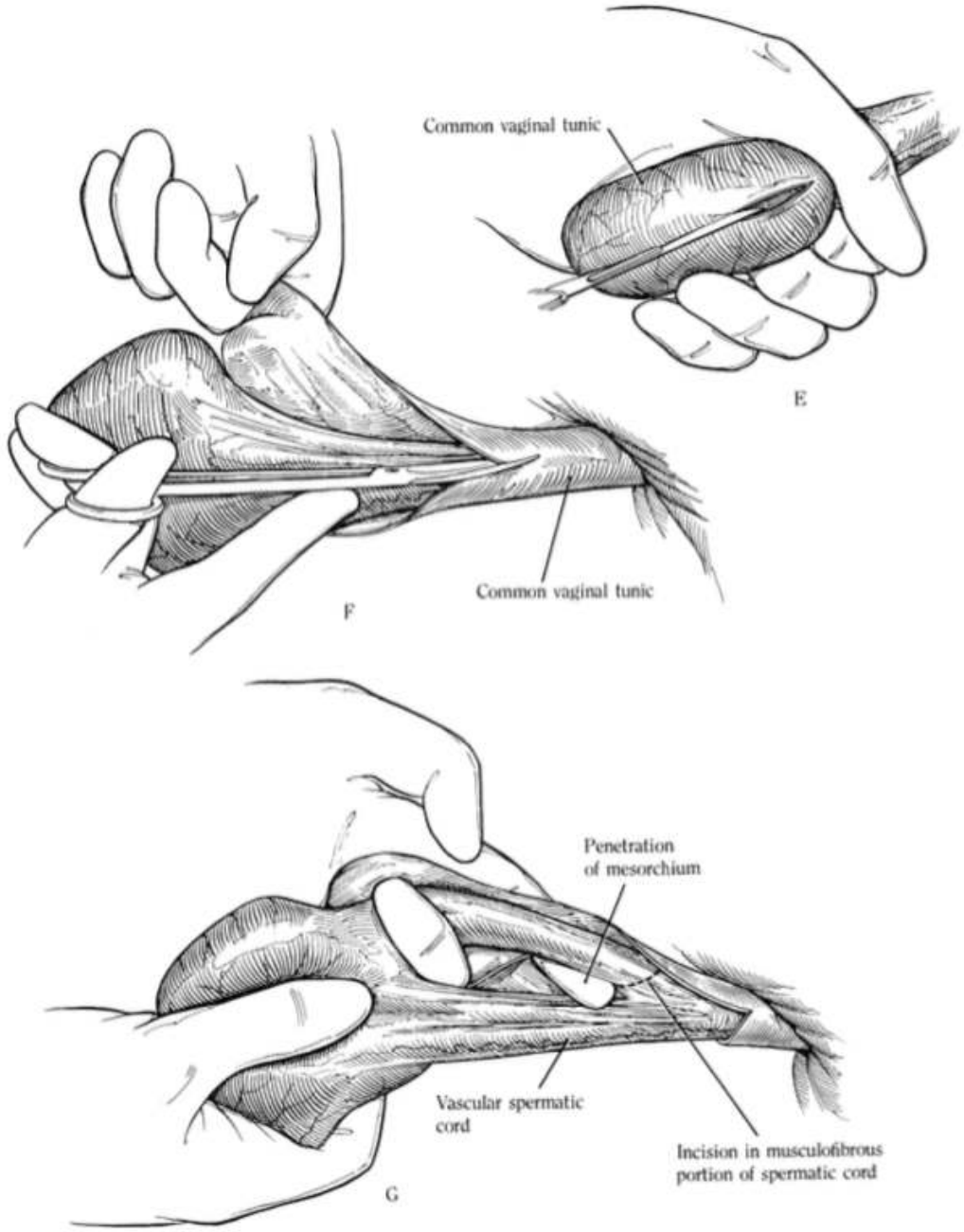
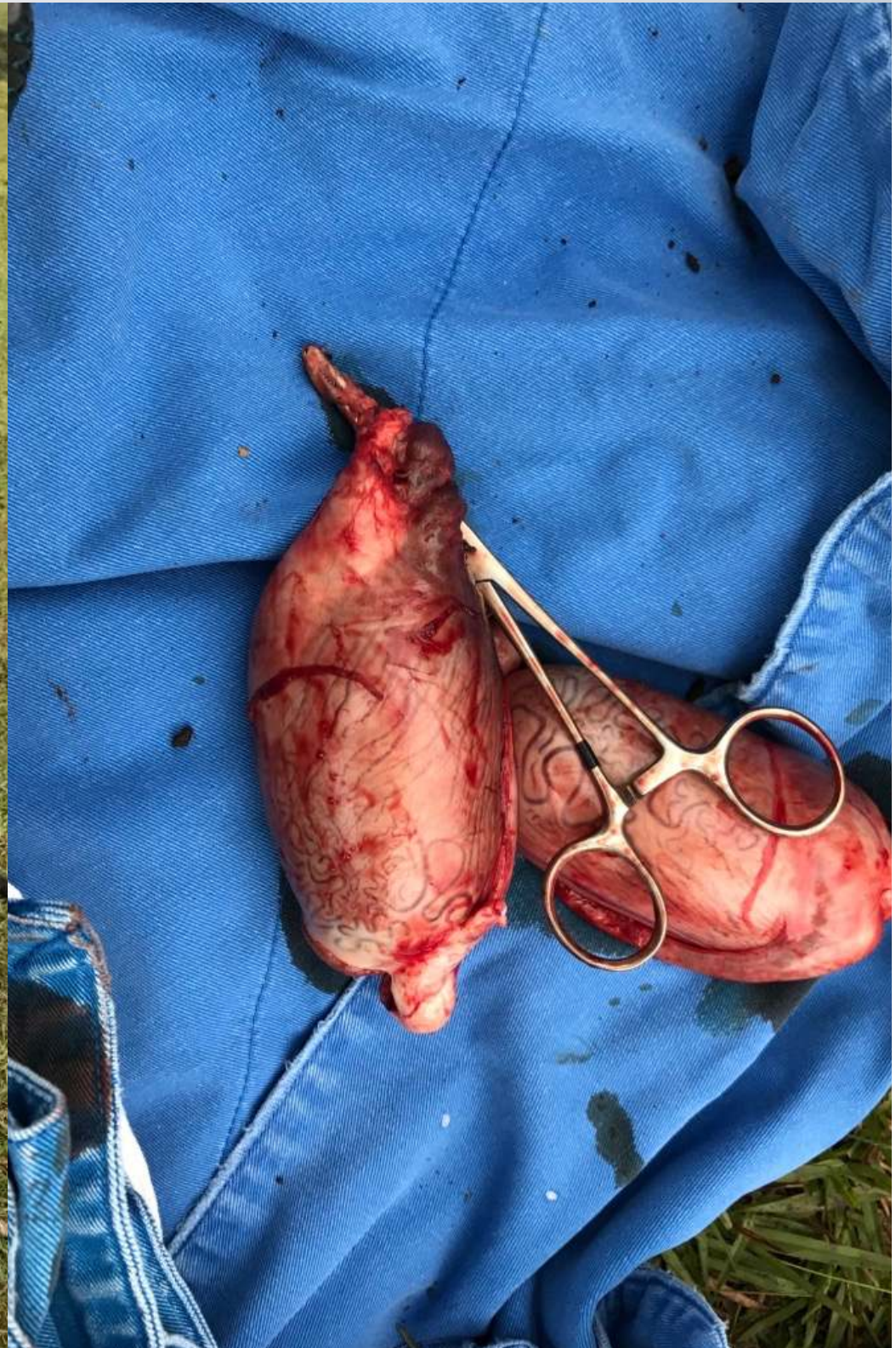


Fig. 10.1. *Continued.*



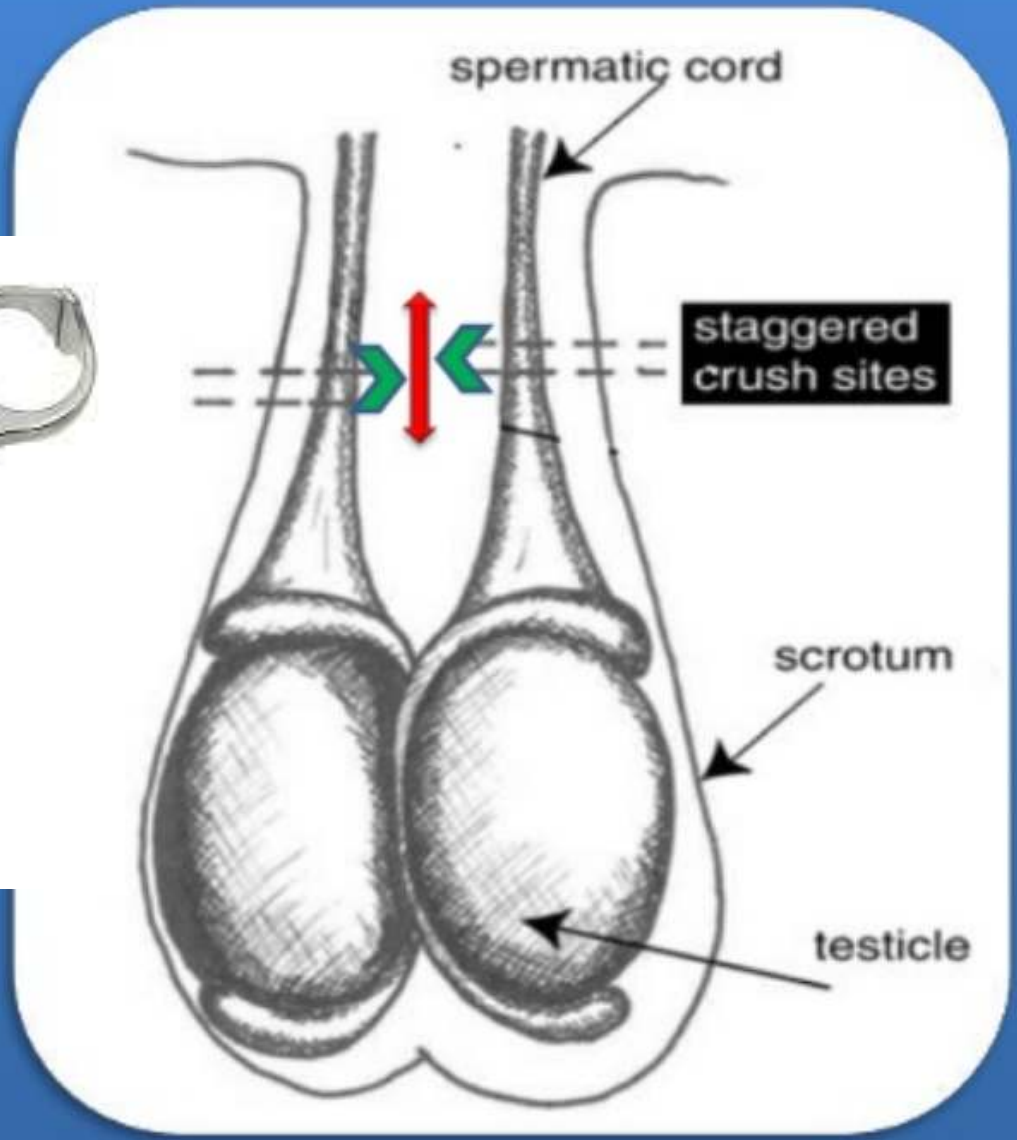




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